

Ceylon Gem Archive

A private gemstone business operating system for managing inventory, costs, provenance, media, buyer interactions, and team access.

Version: Production Stack: React + TypeScript + Supabase Updated: July 2026

— Overview

What this system is and who it's for.

Ceylon Gem Archive (CGA) is a **private, internal tool** for a gemstone trading business. It is not a public marketplace. It's a back-office system where you catalogue every stone you acquire, track its costs and provenance, manage media (photos, videos, certificates, receipts), and selectively share curated views with buyers via secure, expirable links.

The system supports multiple admin users with role-based access (super_admin and admin), managed through an invite-only signup flow.

— Purpose

The problem this solves.

albums	A structured database with search, filtering, and status tracking replaces share pages replace informal photo dumps.
a	Purchase prices, margins, supplier details, and internal notes are never see only what you choose to show.
ustody	Every stone has a timeline of events (mining, cutting, treatment, certifica
bl	Super admins can invite, block, promote, demote, and remove team me inventory.

— Tech Stack

Layer	Technology	Role
Frontend	React 19 + TypeScript	SPA with client-side routing
Styling	Tailwind CSS 3	Utility-first CSS, custom dark theme
Build	Vite 8	Dev server with HMR, production bundler

Layer	Technology	Role
Backend	Supabase (PostgreSQL)	Database, auth, storage, RLS policies
Storage	Supabase Storage	Public bucket for images, videos, certificates, receipts
Hosting	Vercel	SPA deployment with rewrites
Icons	Lucide React	Consistent icon set

CORE MODULES

— Authentication & Roles

How users log in and what they can do.

Standard Supabase email/password auth. No social logins. Session persisted automatically.	Login.tsx
New admins join via invite link from a super_admin. URL: /invite/:token. Token validated and marked used on signup.	InviteSign
admin - Full inventory access: add, edit, view, share. Cannot delete gems. super_admin - Everything admin can do, plus: delete gems, manage templates.	
Super_admin can block an admin. Blocked users see 'account blocked' screen on login. Data remains intact.	AuthConte
Postgres trigger (handle_new_user) auto-creates a profiles row when a	

Key context: AuthContext provides user, profile, isSuperAdmin, and refreshProfile() to all components via React Context.

— Dashboard

The landing page after login.

Four stat cards: Total Gems, Available, Sold, Reserved.	
Three action cards: Add Gemstone, View Archive, Availability Rate (per	
Last 5 gemstones with thumbnail, name, archive number, species, origin, status badge, and weight.	Dashboard

— Gemstone Management

The core inventory system.

Gemstone List

	Full-text search across name, archive number, origin, species. Dropdown
	Switch between card grid (with thumbnails) and compact list view.
	12 items per page with previous/next navigation.
	Multi-select with checkboxes. Selected gems can be shared as a curated list.
Quick View, Edit, Share buttons on each row/card.	Gemstone List

Gemstone Form (Add / Edit)

	Identity: Name, species, variety, stone state (rough/cut&polished) Physical: Weight (carats), dimensions, color, clarity, cut Origin: Country dropdown with common gem origins Treatment: None, Heat Treated, Beryllium, Fracture Filled, etc. Status: Available, sold, reserved, pending Notes: Internal notes (private) and buyer notes (shown on share pages)
	Postgres trigger generates CGA-YYYY-NNNN format, auto-incrementing
	Drag-and-drop photo/video upload. Set primary image, reorder, delete in
	Add, edit, delete history events directly on the form.
	Upload receipt/invoice images in the cost section. media_type: 'receipt'.
Upload certificate photos. media_type: 'certificate'. Shown on detail page.	Gemstone Form

Gemstone Detail

	Large primary image/video with thumbnails. Full-screen lightbox with zoom. Drag-and-drop reordering with sort order saved to database.
	All properties in structured grid: origin, weight, dimensions, color, clarity, etc.
	Purchase price, selling price, cost breakdown, profit margin behind eye icon.
	Chronological event list with inline add/edit. Events: type, date, location, notes.
	Certificate lab, number, uploaded certificate images (clickable). 'Certificate' button.
	Opens share modal to create buyer share link.
Permanently removes gemstone and all associated data. Confirmation required.	Gemstone Deleted

Media Management

Photo, video, certificate, and receipt handling.

Media Type	Purpose	Visible to Buyers?	Storage Path
image	Product photos	Yes	{id}/{filename}
video	Product videos	Yes	{id}/{filename}
certificate	Lab certificate scans	Yes (detail page)	{id}/certificates/{filename}
receipt	Purchase receipts	Never	{id}/receipts/{filename}

Full-screen overlay with keyboard nav (arrows, Escape). Image rotation with re-upload.	Lightbox.
	One primary per gem (star icon). Used as thumbnail. Enforced by database.
	Drag-and-drop reorder on detail page. Persisted to sort_order column.

Cost & Pricing Engine

Multi-currency cost tracking with profit calculation.

	Rough stone price & weight, preforming cost, cutting & polishing, buying other costs. Total auto-calculated.
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Separate currency selectors for purchase, selling, buyer, and sold prices
SGD, JPY, CNY, AED, THB, INR, CHF.

Purchase price: Internal only - total cost

Selling price: Internal asking price

Buyer price: Shown on share pages

Sold price: Recorded when sold

Security rule: purchase_price, selling_price, sold_price, internal_notes, cost fields, supplier info, and receipts must NEVER be exposed on public/buyer-facing pages.

— Provenance & History

Timeline tracking for chain of custody.

Multiple history events per gem displayed chronologically on the detail page

Event type: Free text (Mined, Cut, Certified, etc.)

Date: Optional

Location: Where it occurred

Performed by: Who did it

Notes: Additional details

Edit existing events directly on the detail page. Saved immediately.

[gemstone_id](#)

— Certification

Lab reports and certificate management.

Lab dropdown (GIA, AGL, Gubelin, SSEF, GRS, IGI, AIGS, Lotus Gemology)

Upload scans of physical certificates. Shown as clickable thumbnails on detail page

For uncertified stones. Sets certificate_lab to 'On Request'. Styled message

SHARING & PUBLIC PAGES

— Buyer Share Links

Private, secure pages for individual gems shared with buyers.

Each share generates a unique token. URL: /share/:token. No login required

	Optional expiry. After expiration, shows 'link expired' message. Enforced
	Tracks page views. Incremented on each load. Visible to admins.
	Optional message on the share page (e.g. 'Selected for you, Mr. Patel').
Photos/videos with lightbox, gem properties, buyer price only, buyer notes, provenance timeline, status badge, seller contact info (name, phone, WhatsApp button, extra contacts).	ShareModal

— Collection Sharing

Share multiple gems as a catalog.

	All: Entire inventory filtered by status. Auto-updates as gems change. Curated: Hand-picked gems. Only selected ones appear.
	Choose statuses to include: available, sold, reserved, pending. Multiple
	Name, optional message, optional expiry per collection.
Public page at /catalog/:token. Gem grid with search, status filter, detail modal with media carousel.	Collection

— Public Catalog Pages

What buyers see when they open a shared link.

	Responsive card grid: primary image, name, weight, origin, status. Click
	Media carousel (images + videos) with left/right arrows, property grid, bu
	Buyers can search by name/origin/species and filter by status.
	Collection name, message, seller contact. No CGA branding - white-lab

ADMIN & SETTINGS

— Admin Management

Super admin only. Team member lifecycle.

	Enter email, generate signup link with unique token. Copied to clipboard
	All users with avatar, name, role badge, status badge, phone, join date.
Promote: Admin to Super Admin Demote: Super Admin to Admin Block: Prevents login Unblock: Re-enables login Remove: Permanently deletes profile (hard delete)	AdminManag

— Profile Management

Personal account settings.

Avatar: Upload profile photo Full name: Editable Email: Read-only (Supabase Auth) Phone: Contact number WhatsApp: Shown on buyer pages with WhatsApp button. 'Same as phone' shortcut Additional contacts: Viber, Telegram, WeChat, Signal, email, website, other	ProfilePa
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REFERENCE

— Database Schema

8 tables in the public schema.

Table	Purpose	Key Columns
profiles	User accounts	id, role, status, full_name, phone, whatsapp_number, extra_contacts, avatar_url, invited_by
admin_invites	Signup invite tokens	email, token, used, created_by
gemstones	Core inventory	35+ columns: identity, physical, cost breakdown, pricing, status, notes, timestamps
gemstone_media	Photos/videos/certs/receipts	gemstone_id, url, storage_path, media_type, is_primary, sort_order
buyer_shares	Individual share links	gemstone_id, share_token, caption, expires_at, view_count
gemstone_history	Provenance timeline	gemstone_id, event_type, event_date, location, performed_by, notes
shared_collections	Multi-gem collections	name, share_token, collection_type, status_filter, message, expires_at
shared_collection_items	Gems in curated collections	collection_id, gemstone_id, sort_order

Triggers: handle_new_user (auto-create profile), generate_archive_number (CGA-YYYY-NNNN), update_updated_at (timestamp), enforce_single_primary_media (one primary per gem).

— Security (Row Level Security)

Supabase RLS policies controlling data access.

Table	Authenticated	Anonymous (Buyers)
profiles	Read all. Update own (or super_admin any). Super_admin delete.	None
admin_invites	Full access	Read only (invite validation)
gemstones	Full CRUD (delete: super_admin only)	Read via valid share token or collection
gemstone_media	Full access	Read via valid share or collection
buyer_shares	Full access	Read all. Update view_count
gemstone_history	Full access	None

Table	Authenticated	Anonymous (Buyers)
shared_collections	Full access	Read all. Update view_count
shared_collection_items	Full access	Read all
storage bucket	Upload, update, delete	Read/view all

— Routes

Path	Page	Access
/login	Login	Public
/invite/:token	Invite Signup	Public
/share/:token	Buyer Share Page	Public
/catalog/:token	Catalog Share Page	Public
/	Dashboard	Authenticated
/gemstones	Gemstone List	Authenticated
/gemstones/add	Add Gemstone	Authenticated
/gemstones/:id	Gemstone Detail	Authenticated
/gemstones/:id/edit	Edit Gemstone	Authenticated
/profile	Profile Settings	Authenticated
/admin	Admin Management	Super Admin only

— New Project Setup

Steps to deploy a fresh instance from this codebase.

Create a new project at supabase.com. Note the Project URL and anon

Update .env with:
VITE_SUPABASE_URL=https://xxxxx.supabase.co
VITE_SUPABASE_ANON_KEY=eyJ...

Paste entire contents of `supabase/full-schema.sql` into the SQL Editor and run the functions, RLS policies, and storage bucket.

Sign up through the app. Then promote:
UPDATE profiles SET role = 'super_admin' WHERE id = 'your-uuid';

npm install then npm run dev for local development.

Create a new Vercel project, set environment variables, deploy. vercel.js

Important: The .env file contains Supabase credentials and must never be committed to git.